Glider model

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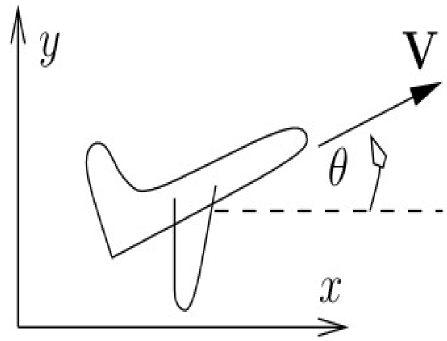
Photo: R.A.X.M. Lubbers

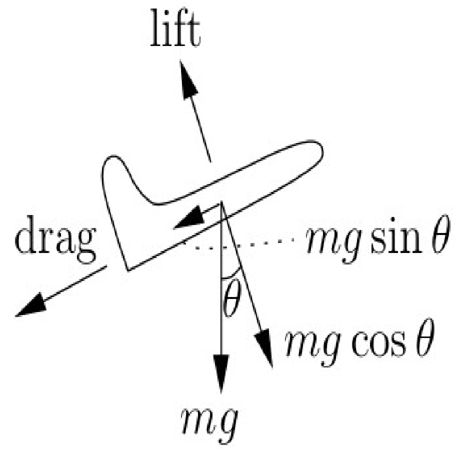
The trajectory of a glider

A simple model to describe the flight of a glider.

Glider model

The glider flies under an angle θ with the horizon and with (scalar) velocity V.





$$
\begin{aligned}
m\frac{dV}{dt} &= -c_D V^2 - mg\sin(\theta) \\
mV\frac{d\theta}{dt} &= +c_L V^2 - mg\cos(\theta)
\end{aligned}
$$

Tasks for the project in Module 3

1. Keep trying to collect quantitative information on gliders and on your glider in particular. If you cannot find that information, make an educated guess. Do not forget to write down where you found the information!
2. Throughout, record your findings. Write down the values you used, store the graphs with descriptive names and write down all conclusions you reached. Store the different versions of the program with descriptive names as well, so you can easily rerun a program later.
3. First take V constant. Use the typical speed of that glider. Implement Euler's method for the differential equation for θ. Use different values for $\theta(0)$ (in radians!). Choose a range for the values of c_L that you want to experiment with. Simulate $\theta(t)$ for the largest value of your range for c_L. (Why the largest c_L?). Experiment with different stepsizes. Estimate the error, using the result with the doubled stepsize. Decide how precise you want to calculate $\theta(t)$, and choose a stepsize accordingly.
4. Then take θ constant and implement Euler's method for the differential equation for V. Use the typical angle the glider flies under on a long undisturbed part of the flight. Choose a few different values for $V(0)$ Now decide on the range of values for c_D that you want to test. Use the highest value of that range for c_D to do the stepsize experiments with. Decide how precise you want to calculate $V(t)$ and choose a stepsize accordingly.

When you have completed the theoretical part of module 3, you can start on the system of differential equations. You could also choose to leave the next points for module 4, but for the glider it is a good idea to start early with the system of differential equations.

5. Choose one or more sets of initial conditions $V(0)$ and $\theta(0)$. Do not choose the equilibrium values for both of the initial conditions $V(0)$ and $\theta(0)$, because nothing will happen then! Implement Euler's method for the system, where both V and θ change in time. If you have decided on different stepsizes for the calculations of the separate $V(t)$ and $\theta(t)$, for the system you need to calculate with the minimum of the two stepsizes. If in doubt check the errors again.
6. When you can simulate the solutions of the system for $V(t)$ and $\theta(t)$, you also want to see what the trajectory of the glider is. The trajectory through the sky is described by the horizontal position $x(t)$ and the vertical position, or height $y(t)$. x and y are described by first-order differential equations of the following form:

$$\begin{aligned}\frac{dx}{dt} &= f_x(V,\theta)\\ \frac{dy}{dt} &= f_y(V,\theta).\end{aligned}$$

   Derive expressions for $f_x(V,\theta)$ and $f_y(V,\theta)$ yourself.
7. Include the two differential equations for x and y into the system in your simulation program (so you solve 4 dependent variables at the same time). Start plotting trajectories. Note that different scales in the figure for the x and y make it difficult to interpret the angles in the trajectory.
8. Calculate the position $(V_{\mathrm{eq}},\theta_{\mathrm{eq}})$ of the equilibrium point analytically. Use a simple relation between a sine and a cosine to find an explicit expression for V_{eq}. In the expression for the θ_{eq} you may approximate $\tan(\theta)\approx\theta$, because it is such a small angle.
9. Find a reasonable values for c_D and c_L. There are different ways for finding a value:
   - Compare the expressions for $(V_{\mathrm{eq}},\theta_{\mathrm{eq}})$ with the values for your glider.
   - Numerical simulation experiments and see whether the resulting trajectories are plausible.
   - Maybe you can find information on c_D or c_L elsewhere. Take care though that you only use constants that apply to the same modelling term. For example, check that the units agree.
   - Calculation of the eigenvalues of the Jacobian matrix in the equilibrium point. If the solution oscillates, determine the period of the oscillations and set c_D or c_L to a value

such that the period of the oscillations is plausible. In that calculation, you can use that for small angles $\cos(\theta) \approx 1$, and $\sin(\theta) \approx \theta$.

Ideally, you use more methods to determine the constants. On the other hand, maybe you want to press on, and focus on other parts of the modelling process. Then you use just one method.

Do communicate well with your teammate, so you make the different choices together. Document every result and decision in your logbook. That will help in the communication between you two, and when you are writing the report later on, you can find all your previous results easily.

After the page for the medicine problem, you will find a page for questions you might have about the project.
